4.3 Further Mechanics

This topic covers impulse, conservation of momentum in two dimensions and circular motion.

It can be studied using applications that relate to, for example, a modern rail transportation system.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

81	understand how to use the equation impulse $= F\Delta t = \Delta p$ (Newton's second law of motion)
82	**CORE PRACTICAL 9: Investigate the relationship between the force exerted on an object and its change of momentum**
83	understand how to apply conservation of linear momentum to problems in two dimensions
84	**CORE PRACTICAL 10: Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top**
85	understand how to determine whether a collision is elastic or inelastic
86	be able to derive and use the equation $E_k = \frac{p^2}{2m}$ for the kinetic energy of a non-relativistic particle
87	be able to express angular displacement in radians and in degrees, and convert between these units
88	understand what is meant by *angular velocity* and be able to use the equations $v = \omega r$ and $T = \frac{2\pi}{\omega}$
89	be able to use vector diagrams to derive the equations for centripetal acceleration $a = \frac{v^2}{r} = r\omega^2$ and understand how to use these equations
90	understand that a resultant force (centripetal force) is required to produce and maintain circular motion
91	be able to use the equations for centripetal force $F = ma = \frac{mv^2}{r} = mr\omega^2$.